Uncertainty-Aware Open-Vocabulary Gaussian Splatting for Changing Scenes

Anonymous Authors

***Abstract*—A robot reusing a semantic map must know how much evidence supports each part and stop trusting content that has changed. We present a probabilistic open-vocabulary field for 3D Gaussian Splatting that serves both needs. Each Gaussian holds a von Mises–Fisher posterior over the direction of its vision-language feature, and each keyframe counts as at most one observation per Gaussian. Its sufficient statistics give an epistemic uncertainty, the weighted number of views, and an aleatoric one, an agreement of the views that does not shrink with scarce evidence. A known feature residual detects change, with a label-free threshold from the map's own mapping pass; a flagged Gaussian is reset and, with depth, rejoins the robust map only once its observed depth matches its position. The two uncertainties separate the two kinds of error: with few views the epistemic one ranks the errors that more views fix (6/6 Replica test scenes), and with many views the aleatoric one ranks those that remain (6/6 Replica, 8/8 ScanNet). On ten real scenes where an object was physically removed, the reset lowers the share of the object's former pixels that still answer its class from 34% (averaging) to 21%. Neither costs accuracy: the field matches averaging the same features, is better calibrated than trained maps, and builds its semantic layer two orders of magnitude faster. As a stale-Gaussian detector, our test fused with depth ranks below residual baselines fused the same way (Gaussian AP 75.0 vs 81.1 and 79.9).**

***Index Terms*—Semantic Scene Understanding, Mapping, Probability and Statistical Methods, RGB-D Perception**

I. Introduction

Robots in homes and offices increasingly query their maps with language [1], [2]. Open-vocabulary 3D Gaussian Splatting (3D-GS) [3] supports such queries by attaching vision-language features [4] to the Gaussians of the map [5]–[8]. A robot that reuses the map needs more than the most likely label: how much evidence stands behind each Gaussian's semantics, whether that evidence agrees, and whether it is still valid. Indoor scenes change, and a map that keeps the old semantics of a moved or removed object answers queries with content that is no longer there.

Several works attach uncertainty to such maps. Bayesian Kernel Inference (BKI) [10], [11] fuses kernel-weighted observations in closed form, and Continuous Semantic Splatting (CSS) [12] brings it to 3D-GS with a Dirichlet posterior over a fixed label set. Other work fuses open-set features on voxels [13], learns an uncertainty [1], [8], or scores the reliability of lifted features [14]. These methods assume a static scene. Change-aware maps [15]–[19] detect moved or removed content by comparing new observations with the map. Two questions fall between these lines of work: how a detected change should enter the uncertainty bookkeeping, and how to set the detection threshold on a new scene without labels.

We answer both inside one Bayesian field. Each Gaussian holds a vMF posterior [20] over the direction of its feature, and a keyframe contributes one weighted observation per Gaussian: the splat-weighted mean of the pixel features it covers. To detect change, we compare all earlier evidence of a Gaussian with the current view. This is the classic feature residual [16], [19], which we do not claim; our part is how its threshold is set without labels, which evidence is reset, and when a Gaussian is trusted again. Geometry stays fixed, so moved content is forgotten at its old place but not inserted at its new one (Fig. 2). Our contributions are:

- **A directional posterior with view-level evidence**, the directional counterpart of Dirichlet semantic mapping. Its epistemic read-out is the weighted number of views; its aleatoric read-out is an agreement of the views that does not shrink with the amount of evidence.
- **Change handling in the Bayesian bookkeeping.** The residual's threshold is a quantile of the map's own mapping-pass scores per history length, a nominal per-view false-alarm rate on static content set without labels (we measure the actual rate). A flagged Gaussian is reset and rejoins the robust map only after a depth check.
- **A predictive that uses both uncertainties**: each pixel's temperature depends on its rendered epistemic and aleatoric uncertainties. We test whether this, or rendering the posterior mean, lowers NLL and ECE against plug-in features at their own fitted temperature.

We evaluate on 8 Replica [21], [22] and 10 ScanNet scenes [23], and on 14 real scenes in which an object was physically removed [24]. We find two capabilities that a single feature or score does not give. First, the two uncertainties separate the errors that more views fix from those they do not on Replica, and the aleatoric one ranks the remaining errors better than the epistemic one on ScanNet. Second, resetting the Gaussians our test flags lowers the share of a removed object's former pixels that still answer its class from 34% to 21%. Neither costs accuracy: the field matches averaging the same features, the strongest baseline we found, and beats trained maps in calibration and build time. Residual baselines fused with depth, however, rank the stale Gaussians better than our test fused with depth.

II. Related Work

Language-embedded 3D-GS. LangSplat [5], Feature 3DGS [6], OpenGaussian [7] and LEGaussians [8] optimize per-Gaussian features against 2D foundation-model features or masks; LEGaussians also learns a per-Gaussian uncertainty.

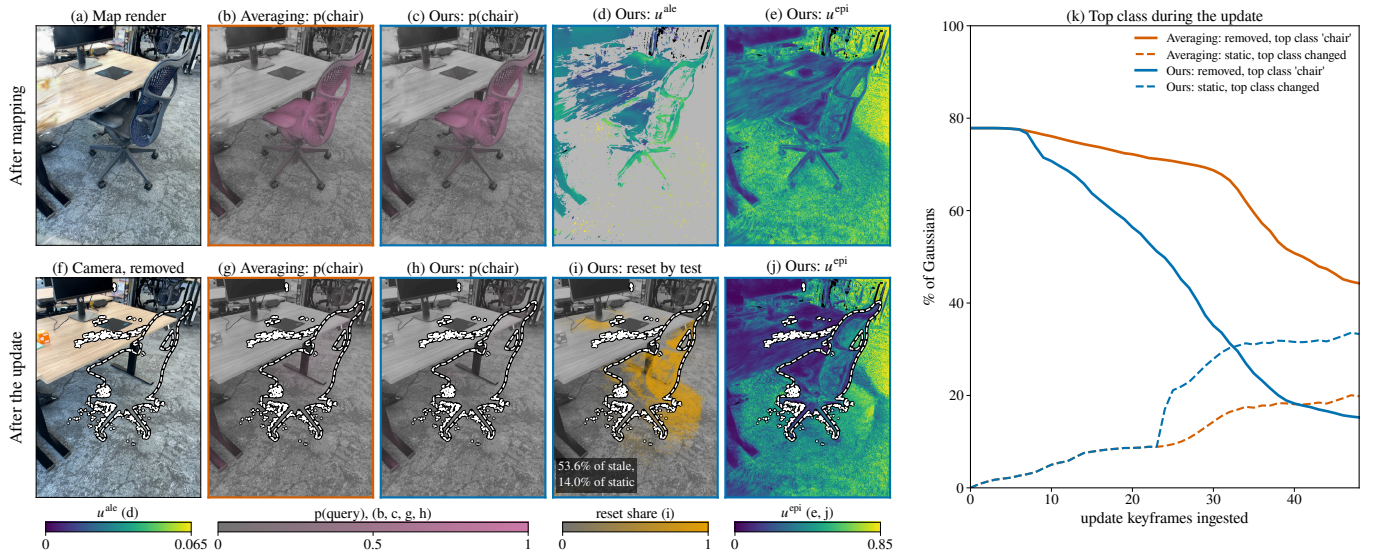


Fig. 1. A semantic 3D-GS map with two uncertainty channels and a test for changed content, on NeRF-Object-Removal validation scene 002 and update keyframe 373 (both picked by rule). Both rows show the same pose; the geometry is fixed, so the removed object's Gaussians are still rendered. Top, after mapping: (a) render; (b, c) p('chair') (query fixed by rule, 101 classes) for averaging and for our field; (d) u^{ale}, whether the views agree (grey: fewer than 3 views); (e) u^{epi}, how much evidence there is. Bottom, after the 48 update keyframes: (f) camera image (dashed: the removed object's Gaussians, ground truth); (g, h) p('chair') after the update; (i) Gaussians our test reset (53.6% of the stale, 14.0% of the static ones); (j) u^{epi}. (k) Over the update: share of the removed object's Gaussians whose top class is 'chair' (solid) and of static Gaussians whose top class changed (dashed).

Training-free methods [9], [25] give each Gaussian the splat-weighted average of its pixel features, which [9] reads from the gradient of one render. The concurrent NormLift [14] averages per-view mean features, factors their norm into within-view and cross-view consistency, and scores reliability by that norm, shrunk by a Kish count [26] of effective views. We reuse the gradient identity and the per-view mean, but our cross-view agreement does not shrink with the evidence, enters a vMF posterior with a calibrated predictive, and handles change.

Probabilistic semantic mapping. BKI [10] updates a conjugate posterior with kernel-tempered observations; Semantic BKI [11] keeps a Dirichlet posterior over classes in voxels, and E2-BKI [27] adds evidential reasoning. CSS [12] brings the Dirichlet model to 3D-GS with the splat weight as the kernel; it separates sparse from conflicting observations over a label set fixed at mapping time and names an open-dictionary conjugate prior as future work. LatentBKI [13] and SLIM-VDB [28] fuse open-set features on voxels under Gaussian models, Bayesian Fields [29] updates per-Gaussian task relevance, and PhysGS [30] splits aleatoric from epistemic variance [31] for physical properties. The vMF distribution is the standard model for unit vectors [20], [32], with a conjugate prior on its mean direction [33]; the language field of LAMP [1] outputs a vMF mean and concentration from a trained network. We train none: we normalize the splat-weight kernel of CSS per view and keep additive vMF statistics per Gaussian (Sec. III-D).

Change detection and long-term maps. 3DGS-CD [15] finds rearranged objects from render-image differences and masks. CL-Splats [16] compares DINOv2 features of renders and images and re-optimizes changed regions. GaussianUpdate [17] uses continual learning with generative replay, and VG-Mapping [18] adds and removes Gaussians in semi-static scenes. MIF [19] updates its memory where a position plus feature-cosine discrepancy exceeds a fixed threshold, and Chu and Kim [34] pick stable objects with a language model. Our score is the same kind of feature residual as theirs; we differ in the label-free threshold and in the response: a detection resets the Gaussian's semantic evidence, and a depth check gates its return.

III. Method

A. Preliminaries

The map holds N Gaussians [3], and its geometry is fixed after mapping. At pixel p, Gaussian n has opacity $\alpha_{p,n}$: its base opacity times its projected 2D Gaussian evaluated at p. With the Gaussians sorted by depth, the splat weight is $w_{p,n} = T_{p,n}\,\alpha_{p,n}$ with the transmittance $T_{p,n} = \prod_{j<n}(1-\alpha_{p,j})$. It is the share of pixel p that Gaussian n explains. A per-Gaussian buffer $\mathbf{f}_n$ renders to $\mathbf{F}_p = \sum_n w_{p,n}\mathbf{f}_n$, and $\alpha_p = \sum_n w_{p,n} \leq 1$ is the rendered opacity. As in CSS [12], $w_{p,n}$ is the BKI kernel [10], [11].

B. Features and Query Space

A dense extractor, LSeg [35], gives unit pixel features $\tilde{\mathbf{z}}_p$ that are cosine-comparable with unit CLIP text embeddings $\tilde{\mathbf{t}}_c$ [4]. An uncentered PCA with orthonormal basis B, fit on mapping frames only (no text, no labels), compresses them to $d = 64$: $\mathbf{z}_p = B^\top\tilde{\mathbf{z}}_p / \|B^\top\tilde{\mathbf{z}}_p\| \in \mathbb{S}^{d-1}$. Texts are projected without renormalization, $\mathbf{t}_c = B^\top\tilde{\mathbf{t}}_c$, so $\boldsymbol{\mu}^\top\mathbf{t}_c = (B\boldsymbol{\mu})^\top\tilde{\mathbf{t}}_c$ is the full-space cosine between the decoded direction $B\boldsymbol{\mu}$ and the text. Renormalizing $\mathbf{t}_c$ would inflate texts with little energy in the image subspace and can change the argmax label.

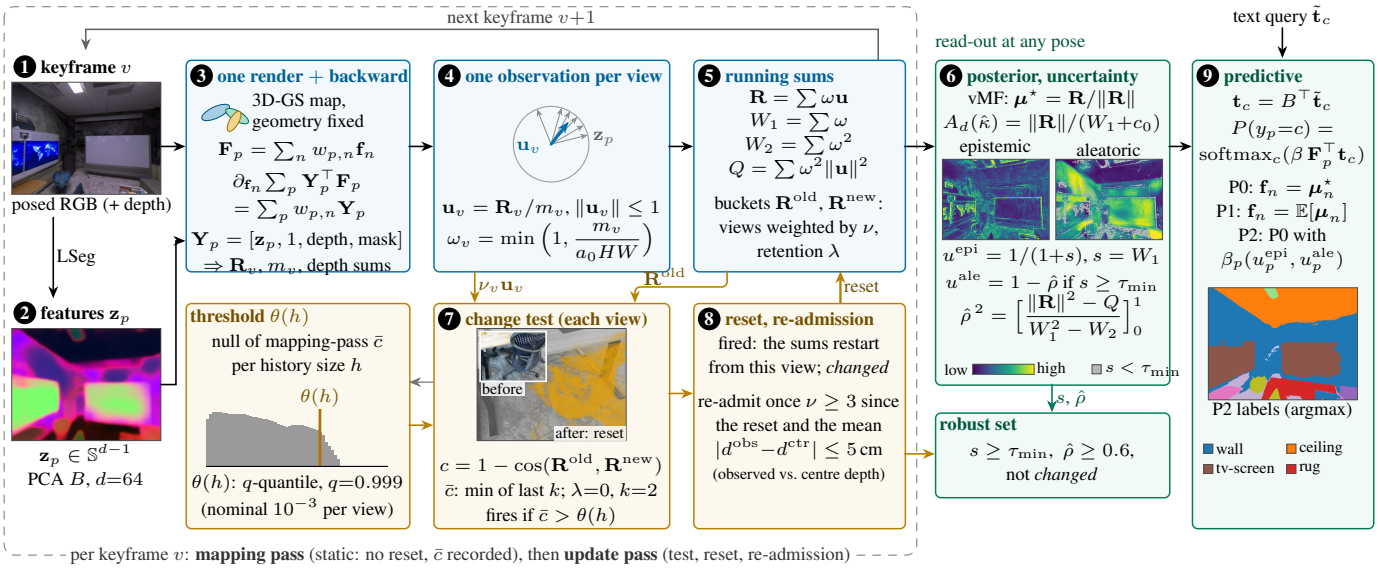


Fig. 2. One keyframe update (mapping and update passes). (1) Posed RGB and, if available, observed depth; (2) LSeg features, compressed by a PCA fit on the mapping frames (first three coordinates as RGB); (3) one render and one backward pass [9] give every Gaussian its per-view sums; (4) each view is one observation $\mathbf{u}_v$ with weight ω_v; (5) four running sums give (6) the vMF posterior, u^{epi} from the support s and u^{ale} from the agreement $\hat{\rho}$ (grey: $s < \tau_{\min}$); (7) a test of all earlier evidence against the current view, with a threshold $\theta(h)$ from the static mapping pass; (8) reset and depth-checked re-admission decide the robust set; (9) a predictive render answers text queries. Index n dropped. Thumbnails: validation scenes Replica office_0 and, for (7) and $\theta(h)$, NeRF-Object-Removal 002 (orange: Gaussians the test reset during the update, over the photo taken after the chair was removed; inset: the map before).

C. View-Level Evidence

Keyframe v gives every Gaussian n the sums $\mathbf{R}_{v,n} = \sum_p w_{p,n}\mathbf{z}_p$ and $m_{v,n} = \sum_p w_{p,n}$. We read them from one render and one backward pass. With geometry and opacity detached, $\mathbf{F}_p$ is linear in the buffer, so the gradient of $\sum_p \mathbf{Y}_p^\top \mathbf{F}_p$ with respect to $\mathbf{f}_n$ is $\sum_p w_{p,n}\mathbf{Y}_p$ for any per-pixel target $\mathbf{Y}_p$. We take this identity from [9] and stack $\mathbf{z}_p$, a ones channel and the observed depth with its validity mask into one target, which gives all per-Gaussian sums without pixel lists; it needs a rasterizer affine in per-Gaussian features, here gsplat [36].

Pixels of one view are strongly correlated (one ViT-L/16 patch of LSeg covers 16×16 pixels), so a pixel count overstates the evidence and grows with the image resolution; we count views. View v gives Gaussian n one observation, the view mean $\mathbf{u}_{v,n} = \mathbf{R}_{v,n}/m_{v,n}$, as in NormLift [14]. Its length $\|\mathbf{u}_{v,n}\| \le 1$ is the coherence inside the view. Its weight $\omega_{v,n} = \min(1, m_{v,n}/(a_0 HW))$ (default $a_0 = 10^{-4}$) does not depend on the resolution, and a view with a smaller footprint counts fractionally. Gaussian n keeps four sums over its views, written without the indices on the right:

$$\mathbf{R}_n = \sum \omega \mathbf{u},\ W_1 = \sum \omega,\ W_2 = \sum \omega^2,\ Q = \sum \omega^2 \|\mathbf{u}\|^2.$$

D. Direction Posterior and Concentration

We use a vMF likelihood $p(\mathbf{z} \mid \boldsymbol{\mu}, \kappa) = C_d(\kappa)\, e^{\kappa \boldsymbol{\mu}^\top \mathbf{z}}$ on $\mathbb{S}^{d-1}$ [20], [32]. Following BKI, we temper the likelihood of pixel p in view v by the kernel $\omega_{v,n} w_{p,n}/m_{v,n}$. A view then carries mass $\omega_{v,n}$ and log-likelihood $\omega_{v,n}\big(\log C_d(\kappa) + \kappa \boldsymbol{\mu}^\top \mathbf{u}_{v,n}\big)$. Given κ_n, the direction posterior is a vMF [33] with natural parameter $\boldsymbol{\eta}_n = \kappa_n \mathbf{R}_n + \kappa_0 \boldsymbol{\mu}_0$, mean direction $\boldsymbol{\mu}_n^\star = \boldsymbol{\eta}_n/\|\boldsymbol{\eta}_n\|$ and concentration $\kappa_n^{\text{post}} = \|\boldsymbol{\eta}_n\|$. The optional text prior is off ($\kappa_0 = 0$), so $\boldsymbol{\mu}_n^\star = \mathbf{R}_n/\|\mathbf{R}_n\|$. For κ_n we use the joint conjugate prior $p(\boldsymbol{\mu}, \kappa) \propto C_d(\kappa)^{c_0} e^{\kappa \mathbf{m}_0^\top \boldsymbol{\mu}}$ with $\mathbf{m}_0 = \mathbf{0}$ (default $c_0 = 1$). The joint posterior is proportional to $C_d(\kappa)^{W_1 + c_0} e^{\kappa \boldsymbol{\mu}^\top \mathbf{R}_n}$, and its MAP concentration solves

$$A_d(\hat{\kappa}_n) = \frac{\|\mathbf{R}_n\|}{W_1 + c_0}, \qquad A_d(\kappa) = \frac{I_{d/2}(\kappa)}{I_{d/2-1}(\kappa)}, \tag{1}$$

where $A_d(\kappa) = \mathbb{E}[\boldsymbol{\mu}^\top \mathbf{z}]$ is the mean cosine of a vMF. We set $\kappa_n = \hat{\kappa}_n$, so $\kappa_n^{\text{post}} = \hat{\kappa}_n \|\mathbf{R}_n\|$; the concentration only sets how sharp the posterior is, never $\boldsymbol{\mu}_n^\star$ or the label. With one-hot features, $\mathbf{R}_n$ equals the count increment of categorical BKI run with our view-level kernel. (CSS [12] uses the pixel kernel $w_{p,n}$, so its counts differ.) The two models share the sufficient statistic and the MAP label but not the distribution, so the vMF model is the directional counterpart of Dirichlet semantic mapping.

E. Epistemic and Aleatoric Uncertainty

Epistemic. The support $s_n = W_1$ is the number of views, each weighted by $\omega \le 1$, and $u_n^{\text{epi}} = 1/(1 + s_n)$. Because $\omega \le 1$, the Kish effective number of views [26] $n_K = W_1^2/W_2$ is never below W_1. It ignores the scale of the weights, so it would count ten small partial views as much as ten full ones.

Aleatoric. Let $\rho_n = \|\mathbb{E}\,\mathbf{u}_{v,n}\|$. It is one when every view shows the same unit feature and near zero when the views disagree. We estimate it by

$$\hat{\rho}_n^2 = \left[\frac{\|\mathbf{R}_n\|^2 - Q}{W_1^2 - W_2}\right]_0^1, \qquad u_n^{\text{ale}} = 1 - \hat{\rho}_n, \tag{2}$$

where $[\cdot]_0^1$ clamps to $[0, 1]$. For independent views with a common mean, $\mathbb{E}\|\mathbf{R}_n\|^2 = (W_1^2 - W_2)\rho_n^2 + \sum_v \omega^2\, \mathbb{E}\|\mathbf{u}_{v,n}\|^2$, so the ratio inside the clamp is unbiased for ρ_n^2 for two or more views; the clamp and the square root leave a small bias at few

views. The aleatoric uncertainty is defined once $s_n \geq \tau_{\min}$ (default 3 views).[1] Two simpler estimates mix the amount of evidence into the dispersion: the mean resultant length $\bar{r}_n = \|\mathbf{R}_n\|/W_1$ is biased upward, by up to about $1/\sqrt{n_K}$, and a prior-shrunk agreement such as $\|\mathbf{R}_n\|/(W_1 + c_0)$, the form of NormLift's reliability [14], is small whenever evidence is scarce. Estimator (2) removes the leading bias of $\bar{r}_n$ and does not shrink, so little evidence and conflicting evidence stay separate (Supp. S4). The shrunk form suits $\hat{\kappa}_n$ in (1), where scarce evidence should broaden the posterior.

F. Change Detection, Reset and Re-admission

We test each Gaussian at each view with a family of two-bucket statistics: an older bucket $\mathbf{R}_n^{\text{old}}$ and a recent bucket $\mathbf{R}_n^{\text{new}}$ split $\mathbf{R}_n$ without overlap. The test counts a view by $\nu_{v,n} = m_{v,n}/\max_{v' \leq v} m_{v',n}$, its share of the largest splat mass the Gaussian has had so far, not by $\omega_{v,n}$: in a dense map most Gaussians never cover $a_0 HW$ pixels in one view, so a test on full views would rarely run (chosen on the removal validation scenes; the static Replica and ScanNet maps predate this choice and count full views, $\nu_{v,n} = \omega_{v,n}$). Each view moves a share $1-\lambda^{\nu}$ of the recent bucket, and of its mass, into the older one, then adds $\nu_{v,n}\mathbf{u}_{v,n}$ to the recent one (retention $\lambda \in [0, 1)$ per unit of ν). The score $c_n = 1 - \cos(\mathbf{R}_n^{\text{old}}, \mathbf{R}_n^{\text{new}})$ is defined once the buckets hold $\tau_{\min}$ views of weight ν and each at least half a view. A change must persist, so $\bar{c}_n$ is the minimum of the last k defined scores.

We chose $\lambda = 0$ and $k = 2$ on a tuning replay not used for testing: in 7-Scenes chess [37], half of the sampled 3D points take the CLIP features of a point over 0.5 m away in their later views. Then $\bar{c}_n$ is the minimum over the last two views t of $1 - \cos\big(\sum_{v<t} \nu_{v,n}\mathbf{u}_{v,n},\ \mathbf{u}_{t,n}\big)$, all earlier evidence against the current view. This member is the classic feature-residual test [16], [19]. Centering the features hurt in tuning and is off.

Threshold from the mapping pass. In tuning, a fixed threshold on $\bar{c}_n$ did not transfer between scenes, so we set it, without labels, from the map's own mapping pass. That sequence is assumed static, so its scores sample the no-change null (no resets happen during mapping). Short histories give noisier scores, so we keep one null histogram per history size, the mass $h_n = W_n^{\text{old}}$ of the older bucket, with bin edges at 4, 6, 10, 20 and 40 views. The threshold $\theta(h)$ is the q-quantile of bin h (default $q = 0.999$), a nominal per-view false-alarm rate of 10^{-3} on static content. A bin with fewer than 200 samples borrows the nearest bin with enough.

Reset and re-admission. When $\bar{c}_n > \theta(h_n)$, Gaussian n forgets: its four sums, buckets and depth record restart from the current view, and it is flagged *changed*. It is re-admitted at the first view at which its weight ν since the reset, the firing view included, is at least 3 and the view-weighted mean of $|d_{v,n}^{\text{obs}} - d_{v,n}^{\text{ctr}}|$ over these views is at most 5 cm. Here $d_{v,n}^{\text{obs}}$ is the observed depth (sensor or estimator, never rendered from the map) averaged over the footprint with the weights $w_{p,n}$, and $d_{v,n}^{\text{ctr}}$ is the camera depth of the Gaussian center. The Gaussians of a removed object see the surface behind them and stay out while it lies more than 5 cm from their centers; without valid depth since the reset they stay out, and without any depth stream the views alone decide. The *robust set* holds the Gaussians with $s_n \geq \tau_{\min}$, $\hat{\rho}_n \geq 0.6$ and no change flag.

[1] One full view is $a_0 HW = 7.68$ pixels of splat mass at our 320×240 resolution. On office_0 / room_2, u^{ale} is defined for 22.4 / 12.2% of the Gaussians but 84.2 / 81.8% of the pixel mass.

G. Predictive Distribution

For pixel p of a novel view, the class probabilities are

$$P(y_p = c) = \text{softmax}_c\left(\beta\, \mathbf{F}_p^{\top}\mathbf{t}_c\right), \qquad \mathbf{F}_p = \textstyle\sum_n w_{p,n}\mathbf{f}_n.$$

The render $\mathbf{F}_p$ is not divided by α_p, so low opacity flattens the distribution. *P0* (plug-in) renders $\mathbf{f}_n = \boldsymbol{\mu}_n^{\star}$. *P1* (posterior) renders the vMF mean $\mathbb{E}[\boldsymbol{\mu}_n] = A_d(\kappa_n^{\text{post}})\,\boldsymbol{\mu}_n^{\star}$, so weakly supported Gaussians pull $\mathbf{F}_p$ toward zero. *P2* (adaptive) keeps the P0 render with a per-pixel temperature $\beta_p = \beta_0 \exp\big(-\gamma_e u_p^{\text{epi}} - \gamma_a u_p^{\text{ale}} - \gamma_u \mathbf{1}[u_p^{\text{ale}}\ \text{undefined}]\big)$, where u_p^{epi} and u_p^{ale} average the defined per-Gaussian values with the weights $w_{p,n}$, and u_p^{ale} is undefined when more than half of α_p comes from Gaussians without it (γ_u gives such pixels their own offset). We fit β (P0, P1), $(\beta_0, \gamma_e, \gamma_a, \gamma_u)$ (P2) and the temperature of every baseline by NLL on the validation split.

IV. Experiments

We ask whether each uncertainty ranks its own error (Fig. 3), whether the reset forgets stale content after a real removal (Table II), whether this costs accuracy or calibration against averaging the same features and trained maps (Table I), and which parts matter (Table III).

A. Setup

Replica. We use the Replica scenes [21] as rendered by Semantic-NeRF [22]: 8 scenes with 101 classes; we ignore unlabeled pixels (id 0). We validate on office_0 and room_2 and test on room_0, room_1 and office_1 to office_4. Each method maps on every 5th frame of Sequence_1 (180 frames; 176 to 180 after dropping near-duplicate poses). We evaluate on every 10th frame of Sequence_2 (90 frames), unseen while mapping. A scene's query vocabulary is the set of classes in its evaluation frames. One ensemble of five prompt templates, chosen on validation, serves all scenes.

ScanNet. We use the ten ScanNet [23] scenes of OpenGaussian [7] and its 19 classes, fixed before any method ran on them, with the Replica templates and settings: scene0000_00 and scene0062_00 validate, the other eight test. We halve the color images to 648×484. Each method maps on every 10th frame (70 to 94% of them kept after dropping near-duplicate and lost poses) and is scored against the filtered 2D labels of every 20th frame from frame 5, which no method maps but which lies 1/6 s from a mapping frame. OpenGaussian scores 3D points, so its numbers are not comparable.

Object removal. Weder et al. [24] recorded each scene twice with ARKit on an iPhone 12 Pro with LiDAR, before and after an object was physically removed, in one coordinate frame (16 scenes). Each method maps on the first sequence and is then updated with the second. We validate on scenes 001 to 004 and test on 005 to 016 except 012 and 016, whose

sequences are offset by 3.5 and 2.5 cm (above our 1 cm limit, fixed before any method ran). A Gaussian is stale if its center lies in the annotated 3D box of the removed object, enlarged by 2 cm, and more than 2 cm from every high-confidence depth point of the second sequence; this leaves out static surfaces inside the box, such as the wall behind a removed bike. No method reads the box. The pixel ground truth is the object mask of the update frames. Re-admission uses the high-confidence LiDAR depth; there is no depth oracle.

Map and features. For each scene we fit one 3D-GS map on the mapping frames with gsplat [36] (degree-0 color, 30k iterations, at most 10^6 Gaussians). Every method uses this fixed geometry and the same LSeg features, compressed to $d = 64$ (Sec. III-B). The coverage floor $\tau_{\text{cov}} = 0.5$ is fixed in advance. We freeze every choice before the test scenes run, and no choice changed after a test run. Besides Tables I and II, the test scenes were also evaluated for the ablations (Table III), the views experiment (Fig. 3) and the later attempts of Supp. S6, all registered before they ran. We report the mean over the test scenes ($N = 6$ for Replica, 8 for ScanNet, 10 for object removal) and, where shown, the population standard deviation.

Segmentation baselines. All baselines share our geometry, features and splat weights. *2D features* apply the extractor to the evaluation frame itself and cover every pixel. *Averaging* is the pixel-weighted mean feature of [9]. *NormLift* [14] keeps that feature and scores reliability $\mathcal{R} = \|\mathbf{f}\| N_{\text{eff}}/(N_{\text{eff}} + 1)$ with Kish effective views N_{eff} over the pixel masses (no KNN voting); we score $1/(1+N_{\text{eff}})$, $1-\mathcal{R}$ and $1-C_{\text{inter}}$, its cross-view consistency. *Feature 3DGS* [6] and *LEGaussians* [8] train the features (Adam, 10^{-3}, at most 30k iterations): the first with an L_1 loss per Gaussian, the second with 128 decoded codes and a learned $u \in [0, 1]$ at the price $\lambda_u \bar{U}$ ($\lambda_u \in \{1, 2, 4\}$, chosen on validation). For reference only, we show CSS [12], a closed-set Dirichlet posterior over per-pixel class votes; it is told the classes of the evaluation frames at mapping time, so we do not rank it (α_0 chosen by validation NLL).

Change baselines. A residual compares an update view with the map rendered at the same pose; we lift it to the Gaussians with the splat weights and, as for our score, require it to persist over $k = 2$ views. *NCC* is one minus the normalized cross-correlation of observed and rendered grey values in small windows. The *depth residual* is the absolute difference between the LiDAR depth of our re-admission check and the map depth. The *feature residual* is one minus the cosine of rendered and observed features, the residual of CL-Splats [16] and the semantic term of MIF [19]. Two rows use an uncertainty as the score: the drop of NormLift's $\mathcal{R}$ during the update and our dispersion $1 - \hat{\rho}$. The *fused* row averages the ranks of our score and the depth residual, with nothing fitted.

B. Metrics

Segmentation and calibration. We report mIoU over the classes in the ground truth, and pixel accuracy. For the predictive we report ECE [38] with 15 equal-width bins on the maximum probability, NLL, the Brier score, and AURC, the area under the risk-coverage curve with pixels ranked by one minus the maximum probability. A labeled pixel with rendered opacity $\alpha_p < \tau_{\text{cov}}$ or a tied top score counts as wrong; an uncovered pixel gets the uniform predictive and ranks as the most uncertain for every uncertainty, and a covered pixel with undefined u^{ale} ranks above every defined value.

Uncertainty ranking. We use the area under the sparsification error [39], $\text{AUSE} = \int_0^1 [\bar{e}_u(f) - \bar{e}_\star(f)]\, df$, where $\bar{e}_u(f)$ is the mean error left after removing the fraction f of pixels with the highest u and $\bar{e}_\star(f)$ uses the error as u; a tie in u takes the mean error of its pixels. For a field built from M of the 180 mapping frames on the full-map geometry, a covered pixel has reducible error ($e_{\text{red}} = 1$) if it is wrong at M frames and right with the full map, and irreducible error ($e_{\text{irr}} = 1$) if it is wrong in both. The epistemic channel should rank e_{red} [31] and the aleatoric channel e_{irr}. AUSE grows with the share of error pixels, which differs between targets, scenes and M, so we divide each scene's AUSE by that of a random ranking of the same errors: 1 is random, 0 the oracle.

Change. We report average precision (AP) and AUROC for stale content at the Gaussian level and at the pixel level, where each row's Gaussian scores are rendered into the update frames and scored on covered pixels against the object mask. A residual row scores a Gaussian by the largest value of its persistent residual over the update, and the uncertainty rows use their value after the update. A Gaussian that a row never scores ranks last, and we report how many there are. The false-alarm rate (FA) is the share of static Gaussians flagged at least once during the update, and recall (Rec.) is the same share for stale Gaussians. Every row sets its threshold as ours does, without labels: the 0.999 quantile of its own scores on the mapping views (Sec. III-F). This targets a nominal per-view false-alarm rate of 10^{-3}, so FA grows with the number of update views that see a Gaussian.

C. Static Scenes: Accuracy, Calibration and Ranking

On Replica (Table I), P0 renders the view-weighted mean of the features that averaging uses and matches it (28.0 vs 27.9 mIoU). Every map row except LEGaussians beats the 2D features, and the posterior mean P1 does not lower NLL or ECE against P0. P2 keeps the NLL of P0 and lowers AURC from 13.8 to 12.2 (on all six scenes) and ECE from 9.9 to 8.7 (on three). It has the lowest mean ECE, Brier and AURC of the open-vocabulary rows, and against averaging it is lower on 3, 5, 6 and 6 of the six scenes in ECE, NLL, Brier and AURC; pooled over all test pixels, however, its ECE is the highest of these rows (7.1 vs 5.7 for averaging), while its Brier score stays the lowest (Supp. S1). Our u^{ale} ranks the errors on par with NormLift's $1 - C_{\text{inter}}$ (AUSE 8.5 vs 8.8), while the learned u of LEGaussians is near a random order (19.7 vs 20.6). Supp. S3 gives per-class IoU and examples.

On ScanNet, our rows are best or tied in every column but tie averaging: P0 raises mIoU from 67.7 to 68.0 (in five of eight scenes, in two by less than 0.01), and P2 lowers the mean ECE from 6.2 to 6.1 and NLL from 0.512 to 0.509 (in three and four scenes), while pooled over pixels averaging has the lower NLL (0.499 vs 0.506). P2 raises AURC over P0 in all eight scenes (5.0 vs 4.4): its temperature, fitted on ScanNet

TABLE I

PARITY CHECK ON THE REPLICA ($N = 6$) AND SCANNET ($N = 8$) TEST SCENES, MEAN. EVERY MAP ROW LIFTS THE SAME LSEG FEATURES, SO THE ROWS DIFFER LITTLE; THE STD OF MIOU OVER SCENES (2.3–4.9, 9.7–10.5) EXCEEDS MOST GAPS. MIOU, ACCURACY, ECE, AURC, AUSE IN %, BRIER ×100, NLL IN NATS. AUSE_u: OF THE ROW'S OWN UNCERTAINTY (LEGAUSSIANS: ITS u; OURS: u^{ale}) AGAINST THE 0/1 ERROR. †OUR RE-IMPLEMENTATION; LEGAUSSIANS λ_u CHOSEN ON VALIDATION (4, 1). **BOLD**/UNDERLINE: BEST/SECOND BEST PER DATASET. CSS: CLOSED SET, NOT RANKED.

	Replica							ScanNet						
Method	mIoU↑	Acc.↑	ECE↓	NLL↓	Brier↓	AURC↓	AUSE_u ↓	mIoU↑	Acc.↑	ECE↓	NLL↓	Brier↓	AURC↓	AUSE_u ↓
2D features, per frame	26.3	74.9	11.8	1.12	40.5	16.3	–	62.5	83.7	8.8	0.64	27.4	7.2	–
Averaging [9]†	27.9	76.7	9.6	**1.00**	36.6	14.0	–	67.7	86.3	6.2	**0.51**	22.3	**4.4**	–
NormLift [14]†, $1 - \mathcal{R}$	same predictive as averaging						14.5	same predictive as averaging						11.4
NormLift†, $1 - C_{\text{inter}}$	same predictive as averaging						8.8	same predictive as averaging						8.3
Feature 3DGS [6]†	27.9	**77.0**	10.0	1.04	37.0	14.6	–	65.0	85.1	7.4	0.57	24.6	5.7	–
LEGaussians [8]†	24.0	74.9	10.6	1.14	40.0	15.8	19.7	62.3	83.6	8.1	0.63	26.8	6.3	10.8
Ours P0 (plug-in)	**28.0**	76.8	9.9	**1.00**	36.4	13.8	**8.5**	**68.0**	**86.5**	6.4	**0.51**	**22.1**	**4.4**	**8.2**
Ours P1 (posterior mean)	27.9	76.7	10.2	1.04	37.5	12.9	**8.5**	67.9	**86.5**	7.0	0.52	22.5	4.8	**8.2**
Ours P2 (adaptive β)	as P0		**8.7**	**1.00**	**36.2**	**12.2**	**8.5**	as P0		**6.1**	**0.51**	**22.1**	5.0	**8.2**
CSS [12]†, closed set	28.5	77.2	5.1	1.01	35.1	11.5	11.2	67.3	86.3	7.0	0.60	22.9	8.9	9.4

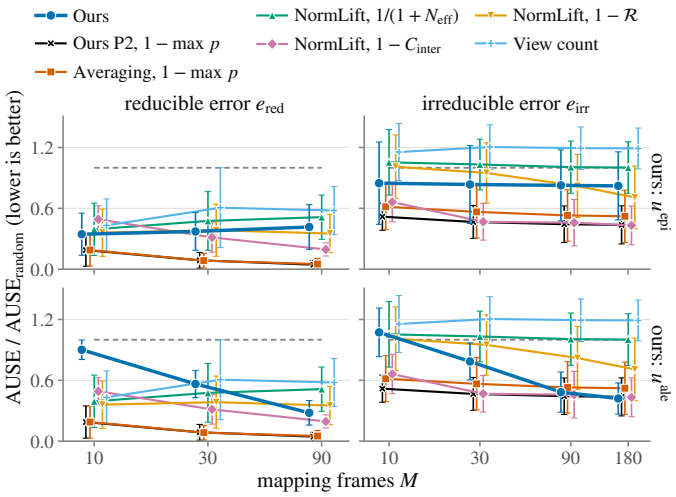


Fig. 3. Does each uncertainty rank its own error? AUSE divided by that of a random ranking of the same errors (dashed; 0 is the oracle) for fields from M of the 180 mapping frames, against the reducible error e_{red} and the irreducible error e_{irr}. Top: our u^{epi}; bottom: our u^{ale}. At $M = 180$ (full map) e_{red} is empty. Replica test, mean ± std, $N = 6$.

validation, grows with uncertainty ($\gamma_e = -0.80$, $\gamma_a = -2.06$; $+0.36$ and $+11.8$ on Replica), the opposite of the intent of Sec. III-G. Our u^{ale} again ranks errors on par with $1 - C_{\text{inter}}$ (8.2 vs 8.3), the trained rows trail averaging by 2.7 and 5.4 mIoU, and closed-set CSS trails our best row in every column. Over the 14 test scenes, a sign test finds no difference between ours and averaging in any metric, while ours has a lower NLL than Feature 3DGS in 13 scenes and beats LEGaussians in every metric (Holm-adjusted $p < 0.05$, Supp. S1); averaging and P0 share this lead, so it comes from training-free lifting, not from the adaptive temperature. Four registered attempts to raise these numbers, each chosen on validation, did not hold: a vote predictive from the same posterior, pooling each Gaussian's statistics over its 3D neighbors, fusing the map with the current frame weighted by our uncertainties, and stronger extractors (RADSeg, ProxyCLIP, Trident; Supp. S6).

Fig. 3 tests whether the two channels separate. With few views, u^{epi} ranks e_{red} better than u^{ale} (0.34 vs 0.90 at $M = 10$, in all six scenes); with many views, u^{ale} ranks e_{irr} better than u^{epi} (0.48 vs 0.83 at $M = 90$, all six scenes; 0.42 vs 0.82 on the full map, five of six). At $M = 10$, u^{ale} is defined on only 4% of the covered pixels. The count-based scores rank e_{irr} close to random or worse (0.82 for u^{epi}, 1.00 for $1/(1 + N_{\text{eff}})$ and 1.19 for the view count, full map): more views do not fix these errors. A total uncertainty, one minus the maximum probability, ranks e_{red} better than either channel; there ours and averaging tie (0.19 at $M = 10$), and on e_{irr} ours is lower at every M (0.44 vs 0.52 on the full map; in 6, 5, 6 and 6 of the six scenes at $M = 10$, 30, 90 and 180). On e_{irr}, u^{ale} is on par with $1 - C_{\text{inter}}$ (0.42 vs 0.43). On the full ScanNet maps, where only e_{irr} exists, u^{ale} again ranks the errors better than u^{epi} in all eight test scenes (0.59 vs 0.87) and on par with $1 - C_{\text{inter}}$ (0.57), but our total uncertainty ranks them worse than that of averaging (0.32 vs 0.26, better in one of eight).

D. Real Object Removal: Stale Content

In Table II, before any reset, our score differs from the feature-residual row only in where it takes the cosine: per Gaussian against all of its earlier evidence, not per pixel against the rendered map. On average, 27% of a test scene's Gaussians are seen by no update view and are not in the table. Our score adds to depth: fused with the depth residual, it raises the Gaussian-level AP and AUROC from 57.4 and 89.3 (depth alone) to 75.0 and 92.3. Fused the same way, however, the feature residual and NormLift's drop rank better (Gaussian AP 81.1 and 79.9; ours + depth higher in 0 and 1 of ten scenes), and NCC and the dispersion rank worse (61.2, 34.2); these rows were added later by a rule fixed in advance (Supp. S2). Alone, our test is not the best ranking: NormLift's reliability drop ranks the stale Gaussians better (AP 63.8 vs 53.4, higher in seven of ten scenes) and is the best unfused row at the pixel level (AP 65.8, AUROC 82.0). At its label-free threshold, our test flags 56.8% of the stale Gaussians at 6.2% false alarms, against 75.6% at 13.7% for the feature residual and 22.8% at 0.6% for the depth residual, and only 4.8% of the stale

TABLE II

STALE-CONTENT DETECTION ON THE OBJECT-REMOVAL TEST SCENES (MEAN ± STD, $N = 10$). METRICS IN %. INPUT: I IMAGE, D LIDAR DEPTH, F FEATURES. †OUR RE-IMPLEMENTATION (FOR [16], [19], OF THE PER-PIXEL FEATURE RESIDUAL ONLY). FA, REC.: AT EACH ROW'S LABEL-FREE THRESHOLD. BOTTOM BLOCK: EACH ROW FUSED WITH THE DEPTH RESIDUAL BY AN EQUAL-WEIGHT RANK AVERAGE (ADDED LATER, BY A PRE-REGISTERED RULE).

		Gaussian level		Pixel level		Operating point	
Method	Input	AP↑	AUROC↑	AP↑	AUROC↑	FA↓	Rec.↑
NCC photometric residual	I	30.4 ± 11.3	75.3 ± 13.5	48.0 ± 14.4	72.5 ± 8.0	12.4 ± 11.1	46.3 ± 28.8
Depth residual	D	57.4 ± 23.3	89.3 ± 5.6	39.7 ± 21.3	67.5 ± 14.2	0.6 ± 0.7	22.8 ± 23.5
Feature residual, $k = 2$ [16], [19]†	F	56.6 ± 23.3	87.6 ± 8.1	57.8 ± 22.2	78.6 ± 11.9	13.7 ± 7.1	**75.6** ± 22.7
NormLift reliability drop [14]†	F	63.8 ± 15.2	91.6 ± 4.6	65.8 ± 14.3	**82.0** ± 9.4	4.1 ± 5.0	41.0 ± 39.6
Dispersion $1 - \hat{\rho}$ (ours)	F	18.4 ± 8.4	53.2 ± 3.6	45.8 ± 23.8	66.4 ± 14.5	**0.5** ± 0.5	8.0 ± 9.0
Ours: test, reset, re-admission	F+D	53.4 ± 22.5	87.2 ± 8.0	57.2 ± 20.3	77.8 ± 11.0	6.2 ± 5.1	56.8 ± 25.5
Ours fused with depth residual	F+D	75.0 ± 12.9	92.3 ± 2.9	61.8 ± 13.8	78.2 ± 11.3	–	–
NCC residual + depth residual	I+D	61.2 ± 17.7	85.3 ± 9.4	56.8 ± 12.0	74.2 ± 12.3	–	–
Feature residual, $k = 2$, + depth residual	F+D	**81.1** ± 9.4	92.8 ± 3.2	64.3 ± 13.4	78.2 ± 12.1	–	–
NormLift reliability drop + depth residual	F+D	79.9 ± 15.1	**93.9** ± 2.4	**66.9** ± 12.2	79.6 ± 12.1	–	–
Dispersion $1 - \hat{\rho}$ + depth residual	F+D	34.2 ± 8.7	82.5 ± 5.3	35.8 ± 22.2	61.7 ± 16.2	–	–

TABLE III

ABLATIONS ON THE TEST SCENES, MEAN OVER $N = 6$ (TOP) AND $N = 10$ (BOTTOM). EACH ROW CHANGES ONE PART OF THE FULL METHOD. AUSE: THE ROW'S u^{epi} (EPI) AND u^{ale} (ALE) AGAINST THE 0/1 ERROR OF THE FULL MAP. AP AT THE GAUSSIAN (G) AND PIXEL (P) LEVEL. KEPT: SHARE OF STALE GAUSSIANS IN THE ROBUST SET AFTER THE UPDATE. ALL METRICS EXCEPT NLL IN %.

Replica test				AUSE↓	
Variant	mIoU↑	ECE↓	NLL↓	epi	ale
Full method	28.0	8.7	1.00	16.4	8.5
Pixels instead of views	27.9	8.1	1.01	16.5	8.8
Uncorrected mean length $\bar{r}$	28.0	8.6	0.99	16.4	8.5
Prior-shrunk agreement	28.0	8.5	1.01	16.4	14.8
Shrunk mean length $\tilde{r}$	28.0	8.7	1.00	16.4	8.5
Centered features	28.0	8.6	0.99	16.4	8.3
Pixel-level variant	27.9	8.1	1.01	16.5	8.8

Object removal, test	AP↑				
Variant	G	P	FA↓	Rec.↑	Kept↓
Full method	53.4	57.2	6.2	56.8	4.8
No persistence ($k = 1$)	47.5	51.3	7.3	56.1	5.3
Pooled null	53.6	56.9	4.6	46.8	6.5
Per-Gaussian max null	53.2	56.5	2.6	32.6	8.6
Re-admission without depth	53.4	57.2	6.2	56.8	9.5
Centered features	48.2	51.1	4.4	43.5	4.6

Gaussians stay in our robust set. The dispersion $1 - \hat{\rho}$ alone does not detect a removal (AP 18.4). As a new measurement, we count the share of the removed object's covered pixels whose top class over the 101 Replica classes is the scene's query: 48% after mapping, 34% after the update for averaging and for our field with the change test off, and 21% for ours, lower than with the test off in nine of ten scenes; the Gaussians that the test misses keep their old class. Letting the stronger fused detectors fire the reset (a later registered attempt) raised the Gaussian AP of the firing score from 53.4 to 79.3 but left more stale answers (26.0 vs 20.7%), mainly because it missed thin objects such as a book, so we keep our test (Supp. S6).

E. Ablations

In the Replica block of Table III, *pixels instead of views* uses $m_{v,n}$ in place of $\omega_{v,n}$, with $u^{\text{epi}} = 1/(1 + \sum_v m_{v,n})$; the agreement rows replace $\hat{\rho}$ (2) by $\bar{r}$, by $\|\mathbf{R}\|/(W_1 + c_0)$, or by $\tilde{r} = \bar{r} - (1 - \bar{r})/(W_1 + 1)$; *centered features* subtract the scene-mean feature before the dispersion and the change test; the *pixel-level variant* turns every part off. Shrinking the agreement raises the aleatoric AUSE from 8.5 to 14.8: it mixes the amount of evidence into the aleatoric channel. The bias correction of $\hat{\rho}$ changes no cell by more than 0.05 on test or validation, so these data do not show its benefit. Counting pixels changes mIoU and AUSE little (27.9; 16.5, 8.8) and lowers the test ECE (8.1 vs 8.7) but raised the validation ECE (5.2 vs 4.9). Centering lowers the aleatoric AUSE a little on both splits (8.3 vs 8.5 test, 4.7 vs 5.0 validation). In the object-removal block, the threshold rows replace the history-conditioned null by one pooled null per scene or by per-Gaussian maxima over the mapping pass. Persistence matters most: with $k = 1$ the AP drops by six points. The history-conditioned null ranks as the others do (AP 53.4 vs 53.6, 53.2) but finds more stale Gaussians (Rec. 56.8 vs 46.8, 32.6) at more false alarms (6.2 vs 4.6, 2.6%). Without the depth check, twice as many stale Gaussians return to the robust set (9.5 vs 4.8%). Centering hurts the change test (AP 48.2).

Run time. A keyframe takes 110 ms on an RTX 4070 Ti (70 ms of it for LSeg; 1.9 GiB peak GPU memory). Building our field on a ScanNet scene takes a median of 5.6 s after LSeg, against 1268 s and 730 s to train the features of Feature 3DGS and LEGaussians (Supp. S5).

V. CONCLUSION

We presented an open-vocabulary semantic field for 3D Gaussian Splatting that counts evidence in views and keeps little and conflicting evidence apart; a known feature residual, thresholded by the map's own mapping pass, drives the reset and depth-checked re-admission of changed Gaussians. The two channels rank the errors that more views fix and those they do not on Replica, and the aleatoric channel ranks the remaining errors on ScanNet; on ten real removals, the reset lowers how often a removed object still answers its query (21% vs 34% of its former pixels). Neither costs accuracy: the field matches averaging the same features and is better

calibrated than trained maps, whose semantic layer takes two orders of magnitude longer to build. As a stale-content ranker, our test fused with depth trails residual baselines fused the same way; combining their scores with our reset, and inserting new geometry where content has moved, are next steps.

Limitations. Geometry is fixed after mapping, so moved content is not inserted at its new place. Fusion cannot fix a class that LSeg confuses in every view. The change threshold assumes a static mapping sequence, and its scores are in-sample (the map and the PCA are fit on those frames), so the false-alarm rate can exceed the nominal rate: on the removal test scenes the measured median is 4.0×10^{-3} per view, four times the nominal (Supp. S2). On ScanNet the fitted P2 temperature grows with uncertainty, so the sign of its coefficients should be constrained or fitted on more data. The 5 cm depth check cannot keep out a thin removed object (a book on a table). All baselines are our re-implementations (LEGaussians on our features) and may differ from the authors' code. On ScanNet, our margins over averaging are smaller than the spread between scenes, and the evaluation frames lie close to mapping frames.

REFERENCES

[1] S. Lee, H. Yu, G. Kim, and S. Choi, "LAMP: Implicit language map for robot navigation," *IEEE Robot. Autom. Lett.*, vol. 10, no. 12, pp. 12 365–12 372, 2025.

[2] K. Nagami *et al.*, "VISTA: Open-vocabulary, task-relevant robot exploration with online semantic Gaussian splatting," *IEEE Robot. Autom. Lett.*, vol. 11, no. 3, pp. 3150–3157, 2026.

[3] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian splatting for real-time radiance field rendering," *ACM Trans. Graph.*, vol. 42, no. 4, pp. 139:1–139:14, 2023.

[4] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 139, 2021, pp. 8748–8763.

[5] M. Qin, W. Li, J. Zhou, H. Wang, and H. Pfister, "LangSplat: 3D language Gaussian splatting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 20 051–20 060.

[6] S. Zhou *et al.*, "Feature 3DGS: Supercharging 3D Gaussian splatting to enable distilled feature fields," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 21 676–21 685.

[7] Y. Wu *et al.*, "OpenGaussian: Towards point-level 3D Gaussian-based open vocabulary understanding," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024, pp. 19 114–19 138.

[8] J.-C. Shi, M. Wang, H.-B. Duan, and S.-H. Guan, "Language embedded 3D Gaussians for open-vocabulary scene understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 5333–5343.

[9] J. Joseph, B. Amrutur, and S. Bhatnagar, "Gradient-weighted feature back-projection: A fast alternative to feature distillation in 3D Gaussian splatting," in *Proc. SIGGRAPH Asia Conf. Papers*, 2025, pp. 1–12.

[10] W. Vega-Brown, M. Doniec, and N. Roy, "Nonparametric Bayesian inference on multivariate exponential families," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 27, 2014.

[11] L. Gan, R. Zhang, J. W. Grizzle, R. M. Eustice, and M. Ghaffari, "Bayesian spatial kernel smoothing for scalable dense semantic mapping," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 790–797, 2020.

[12] J. Wilson *et al.*, "Modeling uncertainty in 3D Gaussian splatting through continuous semantic splatting," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2025, pp. 3284–3290.

[13] J. Wilson *et al.*, "LatentBKI: Open-dictionary continuous mapping in visual-language latent spaces with quantifiable uncertainty," *IEEE Robot. Autom. Lett.*, vol. 10, no. 4, pp. 3102–3109, 2025.

[14] Y. Zang, D. Li, D. Engel, S. Park, and I. Viola, "NormLift: From lifted features to semantic reliability in 3D Gaussian splatting," arXiv:2609.18898, 2026.

[15] Z. Lu, J. Ye, and J. Leonard, "3DGS-CD: 3D Gaussian splatting-based change detection for physical object rearrangement," *IEEE Robot. Autom. Lett.*, vol. 10, no. 3, pp. 2662–2669, 2025.

[16] J. Ackermann *et al.*, "CL-Splats: Continual learning of Gaussian splatting with local optimization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 7808–7817.

[17] L. Zeng *et al.*, "GaussianUpdate: Continual 3D Gaussian splatting update for changing environments," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 25 800–25 809.

[18] Y. He, J. Yu, G. Chen, and H. Zhang, "VG-Mapping: Variation-aware density control for online 3D Gaussian mapping in semi-static scenes," *IEEE Robot. Autom. Lett.*, vol. 11, no. 10, pp. 11 561–11 568, 2026.

[19] P. Jiang, H. Liu, J. Jin, W. Wang, and X. Li, "Learning to evolve: Multimodal interactive fields for robust humanoid navigation in dynamic environments," in *Proc. Robot.: Sci. Syst. (RSS)*, 2026.

[20] K. V. Mardia and P. E. Jupp, *Directional Statistics*. Chichester, U.K.: Wiley, 2000.

[21] J. Straub *et al.*, "The Replica dataset: A digital replica of indoor spaces," arXiv:1906.05797, 2019.

[22] S. Zhi, T. Laidlow, S. Leutenegger, and A. J. Davison, "In-place scene labelling and understanding with implicit scene representation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 15 838–15 847.

[23] A. Dai, A. X. Chang, M. Savva, M. Halber, T. Funkhouser, and M. Nießner, "ScanNet: Richly-annotated 3D reconstructions of indoor scenes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 5828–5839.

[24] S. Weder *et al.*, "Removing objects from neural radiance fields," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 16 528–16 538.

[25] J. Cheng, J.-N. Zaech, L. Van Gool, and D. P. Paudel, "Occam's LGS: An efficient approach for language Gaussian splatting," in *Proc. Brit. Mach. Vis. Conf. (BMVC)*, 2025.

[26] L. Kish, *Survey Sampling*. New York: Wiley, 1965.

[27] J. Kim, M. Jeon, J. Min, K. Kwak, and J. Seo, "E2-BKI: Evidential ellipsoidal Bayesian kernel inference for uncertainty-aware Gaussian semantic mapping," *IEEE Robot. Autom. Lett.*, vol. 11, no. 3, pp. 2378–2385, 2026.

[28] A. Sheppard *et al.*, "SLIM-VDB: A real-time 3D probabilistic semantic mapping framework," *IEEE Robot. Autom. Lett.*, vol. 11, no. 3, pp. 2386–2393, 2026.

[29] D. Maggio and L. Carlone, "Bayesian fields: Task-driven open-set semantic Gaussian splatting," arXiv:2503.05949, 2025.

[30] S. Chopra, J. Liang, G. Seneviratne, and D. Manocha, "PhysGS: Bayesian-inferred Gaussian splatting for physical property estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 18 980–18 990.

[31] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?" in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017.

[32] A. Banerjee, I. S. Dhillon, J. Ghosh, and S. Sra, "Clustering on the unit hypersphere using von Mises-Fisher distributions," *J. Mach. Learn. Res.*, vol. 6, no. 46, pp. 1345–1382, 2005.

[33] K. V. Mardia and S. A. M. El-Atoum, "Bayesian inference for the von Mises-Fisher distribution," *Biometrika*, vol. 63, no. 1, pp. 203–206, 1976.

[34] B.-T.-H. Chu and G.-W. Kim, "Semantic-augmented 3D Gaussian splatting for visual localization in complex indoor environments," *IEEE Robot. Autom. Lett.*, vol. 11, no. 3, pp. 3852–3859, 2026.

[35] B. Li, K. Q. Weinberger, S. Belongie, V. Koltun, and R. Ranftl, "Language-driven semantic segmentation," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[36] V. Ye *et al.*, "gsplat: An open-source library for Gaussian splatting," *J. Mach. Learn. Res.*, vol. 26, no. 34, pp. 1–17, 2025.

[37] B. Glocker, S. Izadi, J. Shotton, and A. Criminisi, "Real-time RGB-D camera relocalization," in *Proc. IEEE Int. Symp. Mixed Augmented Reality (ISMAR)*, 2013, pp. 173–179.

[38] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 70, 2017, pp. 1321–1330.

[39] E. Ilg *et al.*, "Uncertainty estimates and multi-hypotheses networks for optical flow," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 677–693.